

2016

1) Standard free energy change:

The amount of energy released in the conversion of reactants to products under standard conditions is called as standard free energy change. It is the force driving the system towards equilibrium.

It is represented as ΔG°

Its unit is kJ/mol.

2) What is role of Pentose PP in animal tissues?

PPP is a fundamental component of cellular metabolism. The PPP is important to maintain carbon homeostasis, to provide precursors for nucleotide and amino acid biosynthesis to provide reducing molecules for anabolism and to defeat oxidative stress.

3- What do you understand by energy rich compound?

High energy compounds are

also called energy rich compound.

Compounds present in the biological system that when hydrolysed, produce free energy that is greater or equal to that of ATP (ΔG is -7.3 kcal/mol) are termed high energy compounds.

4. What is the ^{role} of pyruvate in a glycolytic pathway? 553

Under aerobic conditions, pyruvate diffuses into mitochondria, where it enters the citric acid cycle and generates equivalents in the form of NADH and FADH₂ -

or from book

Under aerobic conditions, pyruvate is oxidized to acetate (acetyl CoA) which enters the CAC and is oxidized to form CO_2 and H_2O .

The NADH formed by dehydrogenation of G3-P is reoxidized to NAD^+ by passage of its electron to O_2 .

in mitochondrial respiration.

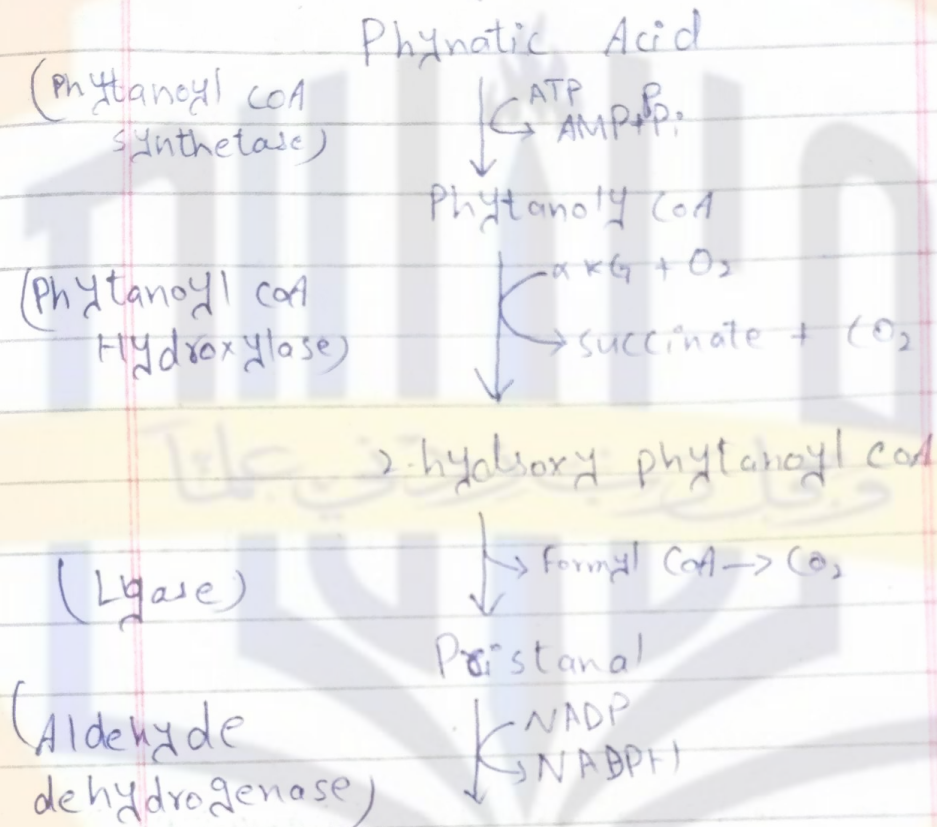
5- Define ^{55%} gluconeogenesis and glycogenesis?

Gluconeogenesis:	Glycogenesis
Organisms need a method for synthesizing glucose from non-carbohydrate precursors. This is accomplished by a pathway called gluconeogenesis. new formation of sugar which converts pyruvate and related 3 and 4 carbon compounds to glucose	The synthesis of glycogen from glucose is called as glycogenesis. It takes place in cytosol and require ATP, besides glucose. Glycogen is stored primarily in muscles ^{cells} and in livers of animals.

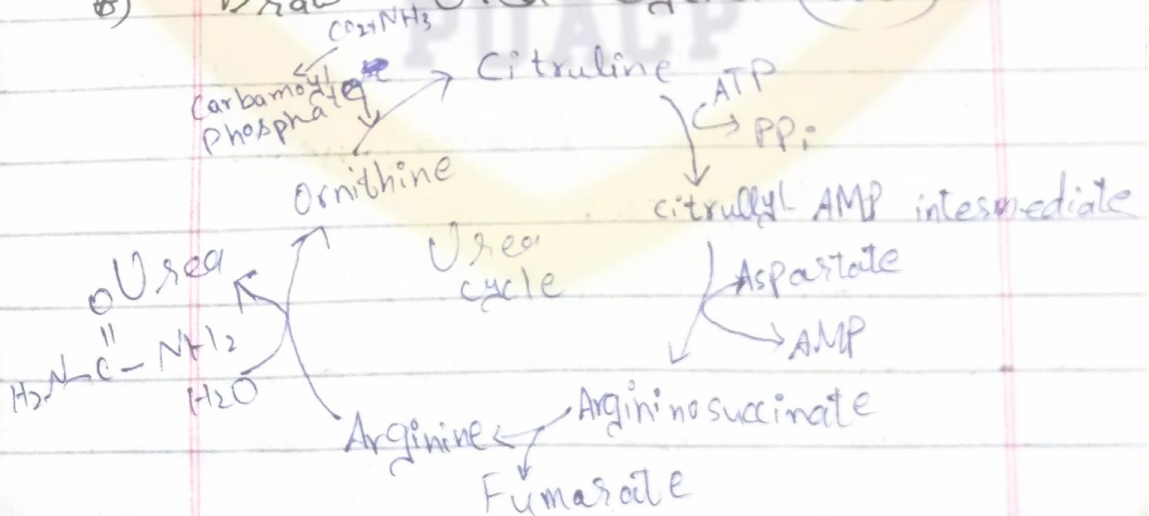
6- Draw pathway for Alpha oxidation of fatty acids.

Alpha oxidation is a process by which certain branched-chain

fatty acids are broken down
by removal of a single carbon
from carboxyl end.



7) Draw Urea Cycle: (685)



8 Write ETC?

ETC is a series of 4 protein complexes that couples redox reactions, creating an electrochemical gradient that leads to the creation of ATP in a complex system named oxidative phosphorylation. It occurs in mitochondria in both cellular respiration and photosynthesis.

The complete cellular process consists of 4 pathways:

- 1) Glycolysis
- 2) Pyruvate oxidation
- 3) Krebs Cycle
- 4) oxidative phosphorylation.

9 How amino acids are catabolized?

The process of AA catabolism starts with a two-step process to remove the amino group. The 2 steps are gluconeogenesis and ketogenesis, followed by oxidative decarboxylation.

AA catabolism is separated into 2 processes.

the breakdown of A.As to carbohydrate precursors, which may lead to the production of glucose (gluconeogenesis) and modifications leading to acetyl CoA, which may lead to contribute to generation of ketone bodies.

10. How glycogen is synthesized?

Glycogen synthesis requires a series of reactions that includes glucose entrance into the cell through transporters, phosphorylation of glucose to glucose 6-phosphate, isomerization to glucose 1-phosphate and formation of uridine 5'-diphosphate glucose, which is direct glucose donor for glycogen synthesis.

Glycogenesis is the process of glycogen synthesis in which glucose molecules are added to chains of glycogen for

storage. It occurs in liver cells and also activated by insulin in response to high glucose levels.

Glucose - 6- Phosphate



Glucose - 1- Phosphate



UDP- Glucose



Glycogen (n+1)

2.2.018

① Point out regulation factors of urea cycle?

Glucagon, insulin and glucocorticoids are major regulators of expression of urea cycle enzymes in liver.

2- How does synthesis of steroids differ from that of peptide hormones?

They differ in relative solubility in plasma-membrane Peptide

hormones consist of 3 or more A.As and are soluble in blood. Steroid hormones are derived

hormones are d) from cholesterol and are insoluble in blood

Amine hormones are derivatives of A.As and some are soluble in blood.

3) Point out rate limiting steps in cholesterol synthesis

The rate limiting step of cholesterol synthesis is the enzyme 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase that synthesizes mevalonate starting from HMG-CoA.

4- What is location of α oxidation of fatty acids

Alpha oxidation of fatty acids occurs in peroxisome as well this metabolic pathway exists to degrade by products of chlorophyll. Phytanic acid is the primary

molecule that requires the enzymes dedicated to α oxidation.

V- Enlist few functions of steroid hormones

Steroid hormones are cyclical chemical compounds made up of rings of carbon atoms that are imp. role in physiological functions, including growth, development, energy metabolism, homeostasis and reproduction.

6- Explain synthesis of steroid hormones?

S.H. are synthesized in mitochondria and smooth E.R. Because they are lipophilic, they cannot be stored in vesicles from which they diffuse easily and therefore synthesized when needed.

7. What are primary and secondary sites for cholesterol synthesis.

About 80% of total cholesterol synthesis takes place in liver

and intestines which are primary sites. The rest being taken place in brain, reproductive organs and adrenal gland. These are secondary sites.

8. Define Allosteric regulation?

Any form of regulation where the regulatory molecule (an activator or inhibitor) binds to an enzyme someplace other than active site is called as allosteric regulation. The place where regulator binds itself is called allosteric site.

Examples:

Glycogen Phosphorylase

Phosphofructokinase

9. Differentiate b/w oxidase and Oxygenase?

Oxidases

Oxygenases

Oxidases are the enzymes that catalyze

the oxidation-reduction reaction ^{by transferring} and thus

The oxygenases are enzymes that catalyze

direct incorporation of oxygen from

the hydrogen from a substrate to oxygen and thus forming water or hydrogen peroxide	molecular oxygen (O_2) to a substrate during the oxidation of a particular substance.
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Example

Cytochrome oxidase
ascorbic acid oxidase

Example

Tryptophan pyrrolase
Carotenase

10- Reducing sugars

A sugar that serves as a reducing agent due to its free aldehyde or ketone functional group in its molecular structure.

Example

Glucose, Fructose, lactose
maltose, except sucrose

Non-reducing sugars

It is a carbohydrate that does not have a free aldehyde or free ketone functional group in its molecular structure. There is free aldehyde or ketone group

Example

Sucrose, Trehalose
Raffinose, Verbascose.

2019

Transketolase

It catalyzes the transfer of a 2 carbon fragment

Transaldolase

It catalyzes a reaction in which a 3 carbon fragment is removed

from a ketose donor | From sedoheptulose
to an aldose acceptor | 7-phosphate and
② It is responsible | condensend with
for generating | G-3-P forming F-6P
reducing equivalent and tetrose erythrose
which is essential, 4 phosphate
for energy trans-
duction and for
generating ribose for
nucleic acid synthesis.

2- How ammonia is transported by
alanine?

Alanine which is formed by trans-
amination of pyruvate gets transported
in the blood to liver, where it
is transaminated by alanine transaminase
to pyruvate. The non-toxic storage
and transport form of ammonia
in liver is glutamine.

3- Write name of sulfur and
hydroxal containing A.As?

Methionine and cysteine are 2
primary sulfur containing amino

acids in mammals - Hydroxal A.As are ^{containing} Threonine, Serine and Tyrosine.

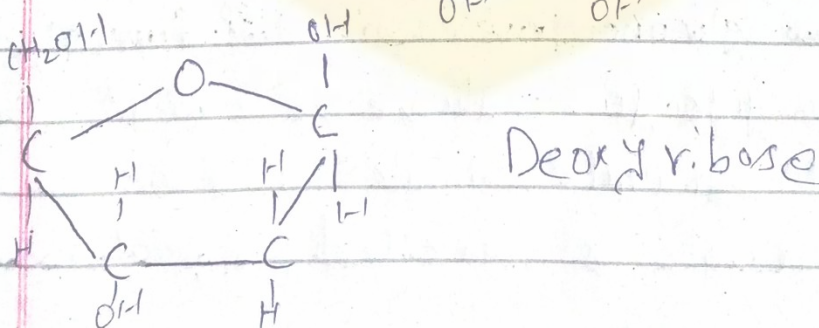
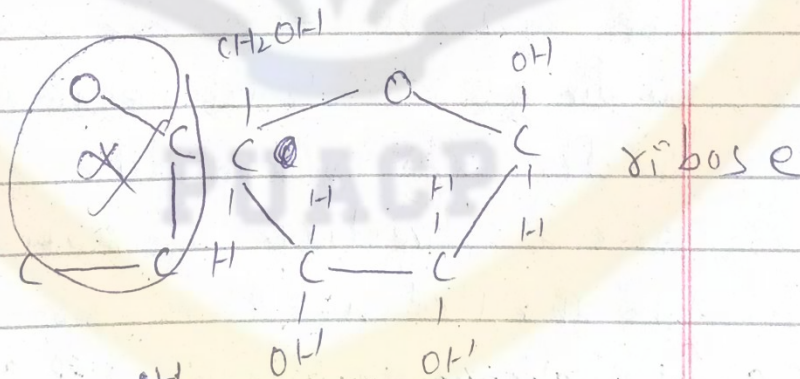
4. What do you know about following Phosphate Ester

A phosphate diester bond occurs when 2 of hydroxal groups in phosphoric acid react with hydroxal groups on other molecules to form 2 ester bonds. These bonds are central to all life on earth.

Glycosidic Bond:

It is formed b/w hemiacetal or hemiketal group of saccharide and hydroxal group of some compound such as alcohol.

5. Structure of ribose and deoxyribose



6- Functions of Lipids:

(i) They are structural compound /

They are part of cell membrane and help control what goes in and out of cells. They help with moving and storing energy, absorbing vitamins and making hormones.

7 Why are phosphorylated intermediate imp?

Several reactions involve the phosphorylation of intermediates, which is imp. not only for production of ATP from ADP, but also as a useful handle on the substrate for enzyme binding to trap intermediate inside cells and to drive pathways in one direction. by making phosphorylation.

8 How gluconeogenesis is diff. from glycolysis?

In glycolysis glucose is converted into pyruvate while in gluconeogenesis pyruvate is converted into glucose.

Glycolysis is a catabolic process of glucose. Hydrolysis needed for energy whereas gluconeogenesis is a glucose production process imp. for maintaining blood glucose levels during starvation.

q. What are fates of pyruvate under anaerobic conditions.

In this process pyruvate is decarboxylated first to form acetaldehyde by the thymine pyrophosphate (TPP) requiring enzyme pyruvate decarboxylase. The resulting product, acetaldehyde, is then reduced by NADH to form ethanol by enzyme ethanol dehydrogenase in a process that forms NAD^+ -

10- Transamination

- The Transamination is the transfer of amino acid group to a keto acid
- Occurs in all cells of body.
- Reversible.

Deamination

- It is the removal of an amino group from an amino acid or other compound.
- Occurs in liver cells.
- Irreversible.

Involves in synthesis of nonessential amino acids | Involves in breakdown of excess proteins.

2020

1- What is Carnitine shuttle?

It represents a mechanism by which long chain fatty acids (which are impermeable to the mitochondrial membranes) are transported into the mitochondrial matrix for the purpose β -oxidation and energy production.

2) How anaplerotic reactions are responsible for regulation of citric acid cycle?

Anaplerotic reactions are metabolic pathways used to replenish oxaloacetate in CAC after it has been consumed. The purpose of these reactions is to maintain adequate levels of ATP so that cellular respiration can carry on uninterrupted.

3- Why ATP is regarded as energy rich compound?

ATP is an excellent energy storage molecule to use as currency due to phosphate groups that link throughout phosphodiester bonds. These bonds are high energy because of the associated electronegative charges exerting a repelling force b/w the phosphate groups.

4. How oxidation of odd numbered fatty?

The only difference is the final product that is produced. In the case of even chain fatty acids, we generate acetyl CoA molecules. But in the case of odd chain fatty acids, we generate acetyl CoA as well as a propionyl CoA molecule.

5. Which amino acids are degraded to pyruvate?

Alanine, cysteine, glycine, serine, threonine, tryptophan.

The pyruvate can then be converted

to either acetyl-CoA or oxaloacetate, thus amino acids catabolized to pyruvate are both ketogenic and glucogenic.

6- Function of Alanine aminotransferase.

(It is also called as glutamate pyruvate transaminase). It is an imp. enzyme in the intermediary metabolism of glucose and protein, catalyzing the reversible transamination b/w alanine and 2-oxoglutarate to form pyruvate and glutamate.

7 Diff. b/w fat and oil

The distinction b/w fats and oils is that fats are solid at room temp. while oil is liquid.

The fats stored in hibernating animal helps them survive. Oil is stored in seeds, which aids in their survival.

8. Why glucose stored in glycogen form?

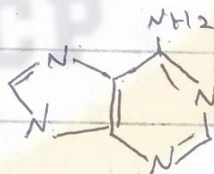
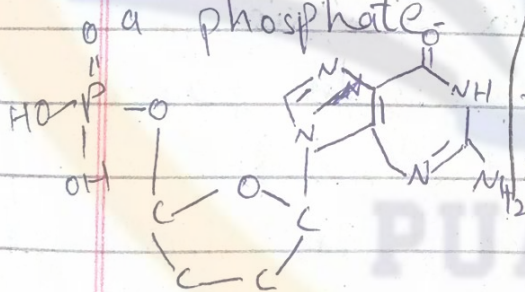
In animal cells, glucose is generally stored in the form of glycogen. This is done to not upset the osmotic balances in cell. Glucose molecules are soluble in H_2O and thus can cause the cell to become hypertonic.

-9. Nucleotide

Nucleoside

It is They are organic molecules composed of a nitrogenous base, a pentose sugar and a phosphate.

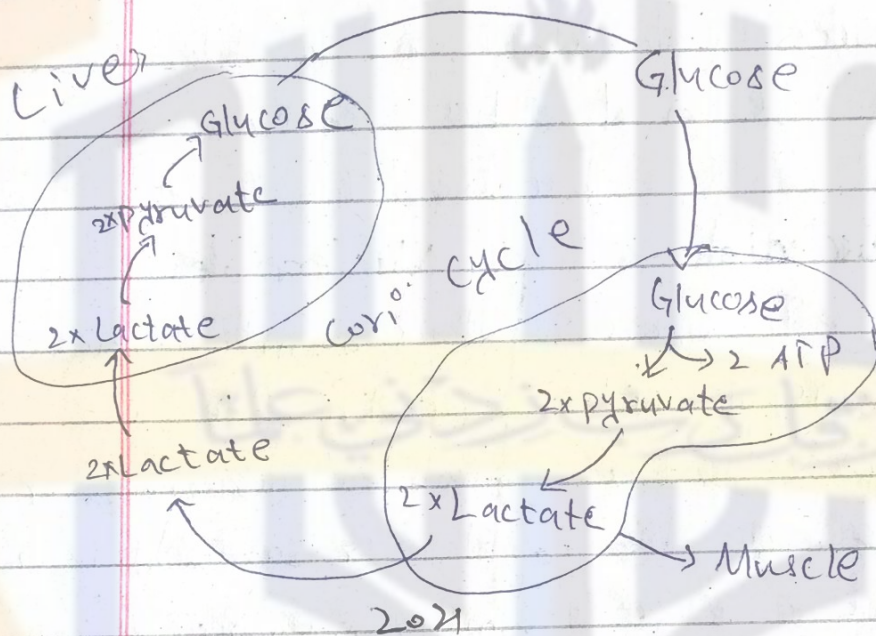
It is a nitrogenous base bond to a pentose sugar ribose or deoxyribose. - It is a pentose sugar and nitrogen containing compound that may form a nucleoside when attached to 5 carbon sugar.



Adenine.

(a) **Corie Cycle:** It is a metabolic pathway in which lactate, produced by anaerobic glycolysis in muscles, is transported to liver and converted to

glucose, which then returns to muscles and is cyclically metabolized back to lactate.



① Feeder Pathway in glycolysis:

Many carbohydrates besides glucose meet their catabolic fate in glycolysis, after being transformed into one of the glycolytic intermediate. The most significant are storage polysaccharides glycogen and starch, either within cell or obtained through diet, the disaccharide

maltose, lactose, trehalose and sucrose; and monosaccharides fructose, mannose and galactose.

In this pathway dietary polysaccharides and disaccharides undergo hydrolysis to monosaccharides. The glycogen inside the cell is degraded by phosphorylation -

2) Regulation of Glycogen Metabolism:

- ① Glycogen metabolism is primarily regulated by hormones (insulin, glucagon, epinephrine).
- ② In skeletal muscles, glycogen metabolism is regulated allosterically. (ATP, AMP, Ca^{2+})

(3) Regulation is based on phosphorylation and dephosphorylation of the key regulatory enzymes which include.

① Glycogen synthase (glycogenesis) active when dephosphorylated

② Glycogen phosphorylase (glycogenolysis) active when phosphorylated.

③ Glycogen phosphorylase kinase:

The conversion of inactive glycogen phosphorylase to active glycogen phosphorylase requires phosphorylation by enzyme phosphorylase kinase.

3) Omega(ω)-Oxidation Pathway:

Omega(ω) oxidation is an alternative pathway to β oxidation that takes place in SER of

kidney and liver in vertebrates.

It is a minor pathway that produces dicarboxylic fatty acids from medium chain fatty acids.

It becomes imp. when pathway for β oxidation is defective.

It has 2 steps.

① Hydroxylation:

In the 1st step a hydroxyl group is introduced at the ω carbon.

② Oxidation: In the 2nd step

the hydroxyl group is oxidized into an aldehyde by alcohol

dehydrogenase, then aldehyde is oxidized to carboxylic acid by aldehyde dehydrogenase. In the end a dicarboxylic fatty acid is produced.

4. Metabolic fates of Amino acids.

Most amino acids are metabolized in liver. In cytosol liver cells amino groups from most amino acids are transferred to α -ketoglutarate to form glutamate, which enters mitochondria and gives up its amino group to form NH_4^+ . In skeletal muscles, excess amino groups are generally transferred to pyruvate to form alanine. The aspartate comes into play in metabolic processes that occur once the amino groups are delivered to liver.

5. Biosynthesis of ketone bodies

6- Types of Prostaglandins:

There are 4 types of Prostaglandins

Type

Function

PGI_2 (Prostacyclin)

Vasodilation

bronchodilation

PGD_2

(Prostaglandin D_2)

critical to develop

of allergic disease

such as asthma

In mammalian organs

large amount of

PGD_2 are found in

brain and in mast cells

PGE_2

(Prostaglandin E_2)

Gastric acid secretion

uterus contraction

lipolysis inhibition

gastric mucous secretion

$\text{PGF}_{1\alpha}$

Uterus contraction

bronchoconstriction